

FastAPI Authentication Failure SOP

Objective

Resolve FastAPI authentication failures caused by JWT validation, expired tokens, invalid secrets, or authorization middleware misconfiguration.

Scope

Applicable to FastAPI-based services, backend APIs, API gateways, DevOps teams, QA teams, and platform teams supporting production, staging, or synthetic incident workflows.

Pre-requisites

- 1. Access to FastAPI application logs, deployment logs, and API gateway or ingress logs.
- 2. Access to monitoring dashboards such as Grafana, Prometheus, APM, or cloud monitoring.
- 3. Permission to view application configuration, environment variables, and deployment status.
- 4. API testing tool such as curl, Postman, Swagger UI, or automated smoke test scripts.
- 5. Escalation contact for backend, DevOps, database, and security teams.

Incident Metadata

Field	Value
Ticket ID	T007
Issue	FastAPI authentication failing
Category	Security
Service	FastAPI
Severity	Medium
Status	Resolved
Description	JWT token validation failing
Expected Resolution	Check token expiration and secret key configuration

Related Log Evidence

Log ID	Timestamp	Level	Message	Error Code
L012	5/1/2026 10:06	WARNING	Authentication failed	API_AUTH
L032	5/1/2026 10:16	ERROR	Authentication failed	API_AUTH

Procedure

- 1. Confirm whether failures are HTTP 401, HTTP 403, or token validation warnings.
- 2. Capture request path, auth header presence, issuer, audience, and token expiry time.
- 3. Review logs for JWT decode errors, expired signature, invalid issuer, or missing scope.
- 4. Validate environment variables for secret key, algorithm, token issuer, and audience.
- 5. Check recent secret rotation, deployment, or identity-provider configuration changes.
- 6. Test with a newly generated valid token and a known expired token.
- 7. Fix middleware, token validation settings, or secret configuration.
- 8. Monitor authentication failure rate and update ticket with validation result.

Common Commands

```
curl -i -X GET http://<host>/health
curl -i -X POST http://<host>/<endpoint> -H "Content-Type: application/json" -d @payload.json
kubectl get pods -n <namespace> | grep fastapi
kubectl logs <fastapi-pod> -n <namespace> --tail=100
kubectl describe pod <fastapi-pod> -n <namespace>
```

Troubleshooting Matrix

Issue	Possible Cause	Resolution
Authentication failed	Expired JWT token	Generate a valid token and check expiry handling
All tokens rejected	Wrong secret or algorithm	Correct auth configuration
Authorized user blocked	Missing scope or role mapping	Fix permission mapping
Random auth failures	Clock skew or token issuer issue	Validate server time and issuer settings

Best Practices

1. Do not log raw JWT tokens.
2. Keep token settings in managed secrets.
3. Add tests for expired and invalid tokens.
4. Alert on sudden increases in HTTP 401 and 403 responses.

References

[FastAPI Docs](#) | [Internal Logs CSV](#) | [Tickets CSV](#) | [Monitoring Dashboards](#) | [API Gateway Logs](#) | [Backend Runbook](#)